

Normal Normal Conjugacy and James Stein Estimator

Contents

1	Bayesian Inference with Normal Likelihoods	2
1.1	Normal Normal Conjugacy	2
1.2	Repeated Measurements of One Parameter	2
2	Multiple Instances of the Same Problem	3
2.1	The Hierarchical Normal Model	3
2.2	Empirical Bayes Estimation	3
2.3	The James-Stein Estimator	4
2.4	Full Bayesian Inference	5
3	Normal Approximation to Binomial and Variance Stabilizing Transformation	5
3.1	The Delta Method	6
3.2	The Arcsine Transformation	6

1 Bayesian Inference with Normal Likelihoods

We will next discuss Bayesian inference with normal likelihoods, and also study some connections to the James-Stein estimator.

1.1 Normal Normal Conjugacy

We will now study Bayesian estimation under the model where data is y and parameter is θ , the likelihood is given by:

$$y \mid \theta \sim N(\theta, \sigma^2)$$

for a fixed $\sigma > 0$. We will look at the case of unknown σ later. The frequentist estimator of θ is the MLE y . For Bayesian inference, we use the normal $N(\mu, \tau^2)$ prior on θ . The basic fact is:

$$\begin{aligned} \theta \sim N(\mu, \tau^2) \text{ and } y \mid \theta \sim N(\theta, \sigma^2) &\implies \theta \mid y \sim N\left(\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2}, \frac{1}{1/\sigma^2 + 1/\tau^2}\right) \\ &\text{and } y \sim N(\mu, \sigma^2 + \tau^2). \end{aligned}$$

The mean of the posterior distribution is thus

$$\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2} = \frac{1/\sigma^2}{1/\sigma^2 + 1/\tau^2}y + \frac{1/\tau^2}{1/\sigma^2 + 1/\tau^2}\mu$$

which is a weighted linear combination of the prior mean μ and the data y . The weights are inversely proportion to the variances of the corresponding normal distributions. For a normal distribution, we use the term “precision” to denote the inverse of variance. Thus the weights of the linear combination above are proportional to the precisions of the prior and the likelihood.

Also note that the precision of the posterior equals the sum of the prior and likelihood precisions.

Remark 1 (Understanding shrinkage). *The posterior mean can also be written as*

$$\mu + \frac{\tau^2}{\sigma^2 + \tau^2}(y - \mu).$$

Thus we start at the prior mean μ and move part of the way toward the observation y . When τ^2 is small relative to σ^2 , we trust the prior more and shrink strongly toward μ . When τ^2 is large relative to σ^2 , we trust the observation more and shrink little. For example, if $y = 2$, $\mu = 0$, and $\sigma^2 = 1$, the posterior means for $\tau^2 = 1/9$, 1 , and 9 are respectively 0.2 , 1 , and 1.8 .

Note that when $\tau^2 \rightarrow \infty$ and μ is a fixed constant, we have $\theta \mid y \sim N(y, \sigma^2)$ (in this case, the posterior mean coincides with the MLE y). Operationally, the $N(\mu, \tau^2)$ prior with fixed μ and $\tau^2 = +\infty$ has the same behavior as the uniform $(-\infty, \infty)$ prior. These are uninformative priors in this problem.

1.2 Repeated Measurements of One Parameter

Often the data will consist of multiple numbers $y_1, \dots, y_n$ with the likelihood being:

$$y_1, \dots, y_n \stackrel{\text{i.i.d}}{\sim} N(\theta, \sigma^2).$$

In this case (for the same prior $\theta \sim N(\mu, \tau^2)$), the posterior is

$$\theta | y_1, \dots, y_n \sim N\left(\frac{n\bar{y}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2}, \frac{1}{n/\sigma^2 + 1/\tau^2}\right).$$

where $\bar{y} := (y_1 + \dots + y_n)/n$. This can be proved directly or use the fact that $\bar{y}$ is sufficient for θ and the data can be reduced to the single number $\bar{y}$ with likelihood $N(\theta, \sigma^2/n)$.

2 Multiple Instances of the Same Problem

2.1 The Hierarchical Normal Model

Consider the problem of estimating $\theta_1, \dots, \theta_N$ from data $y_1, \dots, y_N$ under the likelihood:

$$y_i \stackrel{\text{ind}}{\sim} N(\theta_i, \sigma^2)$$

for a fixed σ^2 . We further assume:

$$\theta_i \stackrel{\text{i.i.d}}{\sim} N(\mu, \tau^2).$$

If we fix μ to be some constant value (say $\mu = 0$) and τ^2 to be very large, then the estimate of θ_i would be equal to y_i .

Remark 2 (Two sources of variation). *Here each observation has its own mean θ_i , unlike the previous subsection, where all observations share one mean θ . The variance σ^2 describes measurement noise around an individual mean, whereas τ^2 describes variation between the individual means. Consequently, the marginal variance of an observation is $\sigma^2 + \tau^2$. Observations from other individuals help us learn the common parameters μ and τ^2 , and thereby help us estimate each θ_i .*

2.2 Empirical Bayes Estimation

Instead of fixing these values, we can treat μ and τ^2 also as unknown parameters and attempt to estimate them from the observed data. Marginalizing θ_i , it is easy to see that

$$y_i | \mu, \tau \stackrel{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2).$$

One can estimate μ and τ from this model by some estimates $\hat{\mu}$ and $\hat{\tau}$. θ_i is then estimated by $\mathbb{E}(\theta_i | y_i, \mu = \hat{\mu}, \tau = \hat{\tau})$. How to obtain $\hat{\mu}$ and $\hat{\tau}$? It is natural to take

$$\hat{\mu} = \bar{y} = \frac{y_1 + \dots + y_N}{N}.$$

The estimate of τ should be based on the sample variance:

$$\sum_{i=1}^N (y_i - \bar{y})^2 \sim (\sigma^2 + \tau^2) \chi_{N-1}^2.$$

The formula for $\mathbb{E}(\theta_i | y_i, \mu, \tau)$ is:

$$\begin{aligned} \mathbb{E}(\theta_i | y_i, \mu, \tau) &= \frac{y_i/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2} \\ &= \mu + \left(1 - \frac{\sigma^2}{\sigma^2 + \tau^2}\right) (y_i - \mu). \end{aligned}$$

So we need to estimate $1/(\sigma^2 + \tau^2)$ as opposed to τ^2 directly. The following fact:

$$\mathbb{E} \left(\frac{1}{\chi_v^2} \right) = \frac{1}{v-2}, \quad v > 2,$$

shows that, for $N > 3$,

$$\begin{aligned} \mathbb{E} \left(\frac{1}{\sum_{i=1}^N (y_i - \bar{y})^2} \right) &= \frac{1}{\sigma^2 + \tau^2} \mathbb{E} \left(\frac{1}{\chi_{N-1}^2} \right) \\ &= \frac{1}{(N-3)(\sigma^2 + \tau^2)}. \end{aligned}$$

Thus an unbiased estimator of $1/(\sigma^2 + \tau^2)$ is

$$\widehat{\frac{1}{\sigma^2 + \tau^2}} = \frac{N-3}{\sum_{i=1}^N (y_i - \bar{y})^2}. \quad (1)$$

The estimate of τ^2 implied by (1) can be nonpositive (depending on the data and σ^2). We shall ignore this however.

2.3 The James-Stein Estimator

The estimates of θ_i are therefore given by:

$$\hat{\theta}_i^{\text{JS}} := \bar{y} + \left(1 - \frac{(N-3)\sigma^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \right) (y_i - \bar{y}).$$

This is the James-Stein estimator. The estimator was introduced by [James and Stein \[1961\]](#) following the celebrated inadmissibility result of [Stein \[1956\]](#).

The James-Stein estimator can thus be seen as an **empirical Bayes** procedure. For a modern discussion of empirical Bayes methods, see [Efron and Hastie \[2016\]](#), Chapters 1–3]. The James-Stein estimator has the following remarkable frequentist property: its risk is uniformly better than the naive estimator $\hat{\theta}_{i,\text{naive}} = y_i$ in mean squared error for all values of $\theta_1, \dots, \theta_N$ (this is true for $N \geq 3$; although we only defined it for $N > 3$, one can create a more naive version of James-Stein replacing $\bar{y}$ by any fixed constant like 0 and this would work also for $N = 3$). This property is beyond the scope of this class.

Remark 3 (Interpreting the risk statement). *The improvement concerns the total risk $\mathbb{E}_\theta \sum_{i=1}^N (\hat{\theta}_i - \theta_i)^2$, not necessarily the risk of each coordinate separately, and not the error on every realized dataset. The displayed estimator, which shrinks toward the estimated grand mean, has the stated strict improvement for $N \geq 4$. For shrinkage toward a fixed constant c , the usual James-Stein formula is*

$$c + \left(1 - \frac{(N-2)\sigma^2}{\sum_{j=1}^N (y_j - c)^2} \right) (y_i - c),$$

and the improvement holds for $N \geq 3$. Thus changing to a fixed target also changes $N-3$ to $N-2$. These frequentist guarantees treat the vector of means as fixed; they do not require a normal prior on that vector.

2.4 Full Bayesian Inference

Let us now present the full Bayes estimate. This requires specification of (uninformative) priors of μ and τ . For μ , we use the uniform $(-\infty, \infty)$ prior. For τ , we use the fact that the marginal distribution of the data given μ, τ is $y_i \mid \mu, \tau \stackrel{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2)$, which is in terms of the parameter $\gamma^2 := \sigma^2 + \tau^2$. For the $N(\mu, \gamma^2)$ model, the standard uninformative prior for γ is

$$\log \gamma \sim \text{uniform}(-\infty, \infty) \quad \text{or} \quad f_\gamma(\gamma) \propto \frac{I\{\gamma > 0\}}{\gamma}.$$

Now we have the additional information that $\gamma > \sigma$ (because $\gamma^2 = \sigma^2 + \tau^2$), so it is natural to modify the above prior as:

$$f_\gamma(\gamma) \propto \frac{I\{\gamma > \sigma\}}{\gamma}.$$

It is easy to check that this prior on γ leads to the following prior on $\tau = \sqrt{\gamma^2 - \sigma^2}$:

$$f_\tau(\tau) \propto \frac{\tau}{\sigma^2 + \tau^2} I\{\tau > 0\}.$$

To recap, we are using the following prior for μ and τ :

$$\mu \sim \text{uniform}(-\infty, \infty) \quad \text{and} \quad f_\tau(\tau) \propto \frac{\tau}{\sigma^2 + \tau^2} I\{\tau > 0\}.$$

We will combine this with the likelihood $y_i \mid \mu, \tau \stackrel{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2)$ to obtain the posterior of μ, τ . It can be checked that:

$$\mu \mid \gamma, y_1, \dots, y_N \sim N(\bar{y}, \gamma^2/N) \quad \text{and} \quad f_{\gamma \mid y_1, \dots, y_N}(\gamma) \propto \gamma^{-N} \exp\left(-\frac{\sum_{i=1}^N (y_i - \bar{y})^2}{2\gamma^2}\right) I\{\gamma > \sigma\}.$$

Remark 4 (Plugging in versus averaging). *Empirical Bayes plugs estimated hyperparameters into the conditional posterior mean. Full Bayes instead averages that conditional mean over the posterior distribution of the hyperparameters:*

$$\mathbb{E}(\theta_i \mid y_1, \dots, y_N) = \mathbb{E}_{\mu, \tau \mid y_1, \dots, y_N} \left[\mu + \frac{\tau^2}{\sigma^2 + \tau^2} (y_i - \mu) \right].$$

Thus both methods borrow information across individuals, while full Bayes also accounts for uncertainty about the hyperparameters.

3 Normal Approximation to Binomial and Variance Stabilizing Transformation

Normal likelihoods are applicable even in Binomial situations, such as in the baseball data analysis example from [Efron and Hastie \[2016, Chapter 1\]](#). Here X_i denotes the number of hits in $n_i = n = 45$ at bats for player i . The natural model is: $X_i \sim \text{Bin}(n_i, p_i)$ with parameter p_i but a normal likelihood (as opposed to Binomial likelihood) can also be used because by invoking the Central Limit Theorem, we can write

$$\frac{X_i}{n_i} \stackrel{\text{approx}}{\sim} N\left(p_i, \frac{p_i(1-p_i)}{n_i}\right).$$

This is because:

$$\sqrt{n} \left(\frac{X_i}{n_i} - p_i \right) \xrightarrow{\text{Law}} N(0, p_i(1 - p_i)) \text{ as } n \rightarrow \infty. \quad (2)$$

Note that the unknown parameter p_i appears in the variance above. If we want to work with normal likelihoods with known variance, a clean way is to use a variance stabilizing transformation. This is justified by the use of Delta method.

3.1 The Delta Method

Theorem 1 (Delta Method). (see, e.g., [Casella and Berger \[2002, Chapter 5\]](#)) If $\sqrt{n}(T_n - p) \xrightarrow{\text{Law}} N(0, \tau^2)$ as $n \rightarrow \infty$, then

$$\sqrt{n}(g(T_n) - g(p)) \xrightarrow{\text{Law}} N(0, \tau^2(g'(p))^2)$$

as $n \rightarrow \infty$ provided $g'(p)$ exists and is non-zero.

Informally, the Delta method states that if T_n has a limiting Normal distribution, then $g(T_n)$ also has a limiting normal distribution and also gives an explicit formula for the asymptotic variance of $g(T_n)$. This is surprising because g can be linear or non-linear. In general, non-linear functions of normal random variables do not have a normal distribution. But the Delta method works because under the assumption that $\sqrt{n}(T_n - p) \xrightarrow{\text{Law}} N(0, \tau^2)$, it follows that T_n converges in probability to p so that T_n will be close to p at least for large n . In a neighborhood of p , the non-linear function g can be approximated by a linear function which means that g effectively behaves like a linear function. Indeed, the Delta method is a consequence of the approximation:

$$g(T_n) - g(p) \approx g'(p) (T_n - p).$$

3.2 The Arcsine Transformation

By the Delta method and (2), we have

$$\sqrt{n_i} (g(X_i/n_i) - g(p_i)) \xrightarrow{\text{Law}} N(0, (g'(p_i))^2 p_i(1 - p_i)).$$

The variance above will not depend on p_i if we choose the function g so that

$$g'(p_i) = \frac{1}{\sqrt{p_i(1 - p_i)}}$$

Solving this for g , we get $g(p_i) = 2 \arcsin(\sqrt{p_i})$. This is a variance stabilizing transformation for the Binomial. Thus by the Delta method, we have

$$2\sqrt{n} \left(\arcsin(\sqrt{X_i/n_i}) - \arcsin(\sqrt{p_i}) \right) \xrightarrow{\text{Law}} N(0, 1).$$

While working with binomial counts, it thus makes sense to transform the observation X_i and parameter p_i into

$$Y_i := 2\sqrt{n_i} \arcsin(\sqrt{X_i/n_i}) \quad \text{and} \quad \theta_i := 2\sqrt{n_i} \arcsin(\sqrt{p_i})$$

respectively and then work with the likelihood

$$Y_i | \theta_i \sim N(\theta_i, 1).$$

Remark 5 (The baseball example and shrinkage). For player i , let p_i denote his underlying probability of getting a hit, and let X_i be his number of hits in his first $n = 45$ at bats. Under the model $X_i \sim \text{Binomial}(45, p_i)$, the natural estimate of p_i is the observed batting average $X_i/45$. However, this estimate can be quite noisy: for a player with $p_i = 0.30$, its standard deviation is approximately $\sqrt{0.30(0.70)/45} = 0.068$. Thus an unusually high or low early batting average may reflect luck as well as ability.

When we observe several players, we can use their collective performance to learn a common level of ability and the amount of variation around it. This suggests shrinking individual batting averages toward a common level, with stronger shrinkage when the observed differences between players can largely be explained by sampling noise.

The arcsine transformation connects this problem to the normal model studied above. Define

$$Y_i = 2\sqrt{45} \arcsin \sqrt{X_i/45}, \quad \theta_i = 2\sqrt{45} \arcsin \sqrt{p_i}.$$

The delta method gives the approximate likelihood $Y_i \mid \theta_i \sim N(\theta_i, 1)$, whose noise variance no longer depends on the unknown batting ability. Under the hierarchical prior $\theta_i \stackrel{\text{iid}}{\sim} N(\mu, \tau^2)$, the posterior mean is

$$\mathbb{E}(\theta_i \mid Y_i, \mu, \tau) = \mu + \frac{\tau^2}{1 + \tau^2}(Y_i - \mu).$$

Here τ^2 describes variation in transformed abilities between players, while the sampling noise has variance one. Empirical Bayes estimates μ and τ^2 from all the players, thereby allowing their data to inform the estimate for each individual player.

For example, with $N > 3$ players, the positive-part James–Stein rule is

$$\hat{\theta}_i = \bar{Y} + \left(1 - \frac{N-3}{\sum_{j=1}^N (Y_j - \bar{Y})^2}\right)_+ (Y_i - \bar{Y}),$$

where $(a)_+ = \max\{a, 0\}$. Notice that N is the number of players, whereas 45 is the number of at bats per player. An estimate on the original batting-average scale is then

$$\hat{p}_i = \sin^2 \left(\frac{\hat{\theta}_i}{2\sqrt{45}} \right).$$

References

- George Casella and Roger L. Berger. *Statistical Inference*. Duxbury, 2 edition, 2002.
- Bradley Efron and Trevor Hastie. *Computer Age Statistical Inference*. Cambridge University Press, 2016.
- Willard James and Charles Stein. Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1:361–379, 1961.
- Charles Stein. Inadmissibility of the usual estimator for the mean of a multivariate normal distribution. *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, 1:197–206, 1956.